

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected.

Centralised cable management also simplifies future moves, adds, and changes.

Case 1: Network Design for a 20-Person Startup

Recommended Topology: Star Topology

- **Hardware Components Required:**
 - 1x Layer 2/Layer 3 Managed Switch (e.g., Cisco Catalyst or Netgear ProSafe) with at least 24 ports.
 - 1x Router/Firewall appliance (e.g., Cisco RV series or pfSense).
 - Wi-Fi Access Point(s).
 - Cat6 UTP (Unshielded Twisted Pair) cables.
- **Design & Implementation:**
 - All end devices (workstations, VoIP phones, printers) connect individually to the central 24-port switch using Cat6 cables.
 - The switch uplinks to the router/firewall, which manages DHCP, NAT (Network Address Translation), and secure internet access.
 - The Wi-Fi Access Point connects to the switch to provide wireless mobility for laptops and mobile devices.
- **Justification (10-Mark Focus):**
 - **Fault Isolation:** If one Cat6 cable fails, only that specific node goes offline. The other 19 employees remain completely unaffected, ensuring business continuity.
 - **Scalability & Management:** Centralized cable management makes it incredibly easy to troubleshoot, move desks, or add new devices as the startup grows.
 - **Performance:** Cat6 supports Gigabit speeds up to 100 meters, which heavily exceeds the requirements of a typical small office floor plan.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access

layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

High-Availability Network for a Bank

Recommended Topology: Full or Partial Mesh Topology

- **Hardware Components Required:**
 - Dual Enterprise-grade Routers (e.g., Cisco ASR or Juniper MX) and Core Switches.
 - Fiber Optic Cables (Single-mode for long distances, Multi-mode OM4 for data centers).
 - Redundant Power Supplies, UPS systems, and RAID storage arrays.
- **Design & Implementation:**
 - **Core Layer:** Configured in a partial mesh where all core switches, routers, and firewalls connect directly to each other.
 - **Distribution/Access Layers:** Use redundant uplinks connected to at least two different core switches.
 - **Connectivity:** BGP-based redundant internet routing is established to maintain external connections.
- **Justification (10-Mark Focus):**
 - **Near-Zero Downtime:** There is no single point of failure. If a link drops, routing protocols (OSPF or BGP) automatically reroute traffic through alternative paths in milliseconds.

- **Data Integrity & Security:** Fiber optics are immune to Electromagnetic Interference (EMI) and support low-latency bandwidth, which is critical for real-time financial transactions.
 - **Regulatory Compliance:** Meets strict financial industry requirements for uptime, backup infrastructure (UPS/RAID), and data availability.
3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Cost-Effective Small Classroom Lab

Recommended Topology: Bus Topology (or Low-Cost Star variation)

- **Hardware Components Required:**
 - *Traditional:* Coaxial cable (RG-58, 50-ohm), T-connectors, BNC connectors, and 50-ohm terminators.
 - *Modern Variation:* 1x Inexpensive 8- or 16-port unmanaged switch, Cat5e UTP patch cables.
- **Design & Implementation:**

- *Traditional Bus:* A single coaxial backbone runs through the room. PCs connect via T-connectors. Terminators cap both ends to absorb signal reflection.
 - *Modern Variation:* PCs connect to a single unmanaged switch. A teacher's PC handles resource sharing natively (Windows/Linux) without needing a dedicated server.
 - **Justification (10-Mark Focus):**
 - **Lowest Hardware Cost:** Eliminates the need for expensive managed hardware, enterprise routers, or dedicated servers, strictly adhering to tight budgets.
 - **Acceptable Trade-offs:** While a single cable break in a strict bus topology disrupts the network, and collisions are more frequent, this is a highly acceptable trade-off for light, supervised usage (like simple web browsing or basic file sharing).
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the

need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

Large University Campus

Recommended Topology: Hybrid Topology (Mesh/Ring Core + Star Building)

- **Hardware Components Required:**
 - Core Routers and Multilayer Switches.
 - High-speed Fiber Optic Cables (Single-mode, 10 Gbps or 40 Gbps).
 - Distribution Switches, Access Switches with PoE (Power over Ethernet).
 - Cat6 cables and Wireless Access Points.
- **Design & Implementation:**
 - **Core Layer (Campus Backbone):** Uses a fiber-based Ring or Partial Mesh to interconnect core routers.
 - **Distribution Layer (Building Level):** Each building has a distribution switch connected to the core via dual redundant fiber uplinks.
 - **Access Layer (Floor/Room Level):** Traditional Star topology connecting end devices via Cat6. APs are powered directly via PoE.
- **Justification (10-Mark Focus):**
 - **Scalability:** Expanding the campus requires zero disruption. Adding a new dorm or lab simply means running a new fiber link to a new distribution switch.
 - **Tiered Redundancy:** The Mesh/Ring core ensures no single fiber cut isolates an entire building, while the Star access layer keeps endpoint troubleshooting simple.
 - **Optimization:** PoE eliminates the need for separate electrical wiring for campus-wide Wi-Fi APs, saving massive installation costs.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are

then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

Multi-Department Corporate Network

Recommended Topology: Hierarchical / Extended Star with VLANs

- **Hardware Components Required:**
 - Dedicated Layer 2 Access Switches (for HR, Finance, Sales, IT).
 - Central Layer 3 Switch or Router (Core Layer).
 - Edge Router with integrated Firewall.
- **Design & Implementation:**
 - Each department gets its own Layer 2 switch (Access Layer).
 - These switches uplink to a central Layer 3 switch or router that handles Inter-VLAN routing.
 - Virtual LANs (VLANs) are configured to logically segregate the departments, mapped to proper IP subnetting.
- **Justification (10-Mark Focus):**
 - **Enhanced Security:** VLANs isolate department traffic. For example, Sales cannot sniff packets or access internal servers dedicated to Finance or HR unless explicit routing rules allow it.
 - **Traffic Optimization:** Broadcast domains are heavily reduced. A broadcast storm in the IT department will not flood the HR network, vastly improving overall network performance.

- **External Protection:** The edge router with firewall functionality acts as a secure gateway, shielding internal segregated networks from external internet threats

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation	5%	Clear,	Good	Basic	Poor or no

& Presentation		professional, and well- documented	documentation	documentation	documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution